MediWorks

# AL550

## Optical Biometer

*Accurate Biometric Measurement*
*for Cataract Surgery*





Optical Biometer AL550

# Do you know the causes of blindness and moderate to severe visual impairment in people aged 50 and above?

- With the global aging population, **cataract, uncorrected refractive error, glaucoma, age-related macular degeneration and diabetic retinopathy** have become the main causes of blindness and moderate to severe visual impairment (MSVI) in people aged 50 and above.

- **Cataract remains the 1st or 2nd leading cause of blindness and MSVI** in all world regions so far. However, the World Health Assembly (WHA) defines cataract as one of the avoidable causes of vision impairment because it is so effectively treated with surgery.

| | Cataract | Undercorrected refractive error | Glaucoma | Age-related macular degeneration | Diabetic retinopathy | Residual causes of vision loss |
|---|---|---|---|---|---|---|
| **Blindness** | | | | | | |
| Global | **45.4% (41.7–49.0)** | 6·6% (5.6–7.8) | 11% (9.3–12.8) | 5.6% (4.3–6.7) | 2.5% (1.7–3.7) | 28.9% (26.5–31.5) |
| Central Europe, eastern Europe, and central Asia | **22.4% (19.1–25.9)** | 1.4% (1.2–1.8) | 14.8% (12.3–17.3) | 5.2% (3.8–6.7) | 1.0% (0.7–1.5) | 55.1% (51.0–59.6) |
| High income | **17.5% (14.9–20.5)** | 2.1% (1.7–2.5) | 28.2% (24.0–32.3) | 21.6% (17.5–26.1) | 6.2% (4.2–8.8) | 24.5% (21.6–27.8) |
| Latin America and Caribbean | **35.4% (31.4–39.8)** | 4.3% (3.6–5.0) | 11.8% (9.7–13.9) | 2.5% (1.8–3.2) | 6.8% (4.5–9.6) | 39.2% (35.5–43.4) |
| North Africa and Middle East | **33.6% (29.4–38.0)** | 3.3% (2.7–3.8) | 21.1% (17.7–24.5) | 8.3% (6.2–10.5) | 2.3% (1.5–3.6) | 31.5% (27.8–36.0) |
| South Asia | **63.1% (59.4–66.8)** | 9.4% (7.9–10.9) | 6.4% (5.2–7.7) | 3.0% (2.1–4.0) | 1.4% (0.9–2.1) | 16.8% (15.0–18.8) |
| Southeast Asia, east Asia, and Oceania | **48.3% (44.5–52.1)** | 7.3% (6.2–8.5) | 6.7% (5.5–7.9) | 4.2% (3.0–5.4) | 2.0% (1.3–3.0) | 31.6% (29.0–34.3) |
| Sub-Saharan Africa | **39.8% (35.9–43.9)** | 3.1% (2.6–3.6) | 17.8% (15.1–20.6) | 4.0% (3.0–5.1) | 0.9% (0.6–1.4) | 34.4% (31.4–37.7) |
| **Moderate and severe vision impairment** | | | | | | |
| Global | **38.9% (35.6–42.4)** | 41% (38.0–44.1) | 2.1% (1.7–2.5) | 3.0% (2.5–3.5) | 1.4% (1.0–2.0) | 13.6% (12.3–15.2) |
| Central Europe, eastern Europe, and central Asia | **21.4% (18.8–24.2)** | 45.3% (42.0–48.6) | 1.5% (1.2–1.8) | 1.6% (1.3–1.9) | 1.0% (0.7–1.3) | 29.3% (26.2–32.5) |
| High income | **33.5% (29.8–37.2)** | 44.5% (41.1–48.0) | 2.5% (2.0–3.1) | 3.2% (2.7–3.8) | 1.9% (1.3–2.5) | 14.5% (12.7–16.8) |
| Latin America and Caribbean | **31.8% (28.4–35.3)** | 40.2% (37.1–43.4) | 3.6% (3.0–4.4) | 2.4% (2.0–2.8) | 2.8% (2.0–3.8) | 19.2% (17.0–21.8) |
| North Africa and Middle East | **44.6% (40.6–48.8)** | 36.3% (32.7–39.8) | 2.9% (2.3–3.6) | 4.2% (3.5–5.0) | 3.2% (2.3–4.3) | 8.9% (7.6–10.6) |
| South Asia | **41.3% (37.8–45)** | 44.9% (41.3–48.1) | 1.5% (1.2–1.8) | 1.8% (1.5–2.2) | 0.6% (0.4–0.8) | 10% (8.9–11.2) |
| Southeast Asia, east Asia, and Oceania | **43.8% (40.4–47.2)** | 36.6% (33.7–39.7) | 1.9% (1.5–2.3) | 4.3% (3.6–5.0) | 1.7% (1.2–2.3) | 11.8% (10.8–13.0) |
| Sub-Saharan Africa | **43.3% (39.8–46.8)** | 26.7% (24.4–29.0) | 3.9% (3.2–4.8) | 4.2% (3.5–4.9) | 1.2% (0.9–1.6) | 20.8% (18.7–23.0) |

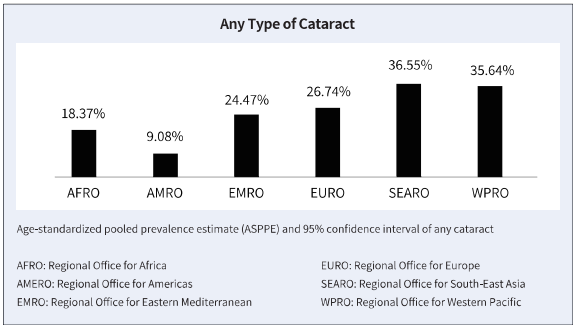
Data in parentheses are 95% uncertainty intervals.

**Relative percentage contribution of each cause to age-standardised prevalence of blindness and moderate and severe vision impairment by superregion in 2020 in adults aged 50 years and older**

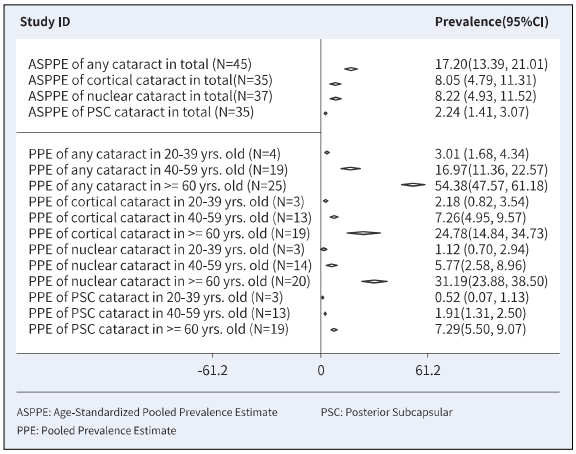
Data source: GBD 2019 Blindness and Vision Impairment Collaborators.Causes of blindness and vision impairment in 2020 and trends over 30 years, and prevalence of avoidable blindness in relation to VISION 2020: the Right to Sight: an analysis for the Global Burden of Disease Study. Lancet Glob Health 2021; 9: e144–60.

# What's the prevalence of cataract?

- A systematic review and meta-analysis demonstrated the Age-standardized Pooled Prevalence Estimate (ASPPE) of cataract was 17.20%, but it was significantly different in the six geographical regions. The highest prevalence was 36.55% in the SEARO region and the lowest prevalence was 9.08% in the AMRO region.




- It also showed the Pooled Prevalence Estimate (PPE) of any cataract in the 20-39-year, 40-59-year, and over-60-year age groups was 3.01 (95% CI: 1.68–4.34), 16.97% (95% CI: 11.36–22.57), and 54.38% (95% CI: 47.57–61.18), respectively.




- It' s crucial to note that due to the global trend of population aging, the ASPPE of cataract is anticipated to increase in the future. Although effective surgical interventions are available, cataract - being the first leading cause of blindness worldwide - warrants heightened attention from eye-care professionals.

Data source: Hassan Hashemi, et al. Eye (Lond). Global and regional prevalence of age-related cataract: a comprehensive systematic review and meta-analysis. 2020 Aug; 34(8): 1357-1370.

# How can AL550 Optical Biometer help you on the cataract surgery for your patients?

## Achieve High Success Measurement Rate in Cataract patients:

- The wavelength of AL550 features high penetrability through ocular tissues, thereby improving the measurement success rate in cataract patients.

- AL550 offers multiple clinical applications, such as **cataract surgery, keratoconus screening, wavefront aberration analysis**, providing a comprehensive basis for perioperative eye health evaluation in cataract patients.

## Enable Precise Prediction of IOL Power:

- AL550 provides accurate biometric measurements, such as **axial length, corneal curvature** and **anterior chamber depth**, thereby reducing errors in calculating IOL power for cataract patients.

- AL550 offers multiple new-generation IOL calculation formulas, including **Pearl-DGS** and **Jin**, whose accuracy has been validated to be comparable to Barrett Universal II.

## Provide an Easy-to-operate and Patient-friendly Examination Process:

- AL550, with selectable fully automatic or manual operation, voice prompt and rapid capture, makes it easier for eye-care professionals to examine cataract patients.

- AL550 provides elderly patients an excellent feeling of comfort during the examination, enabling them to cooperate with the measurement at ease.



# Measurement Principle of AL550

## OLCR (Optical Low-Coherence Reflectometry) Measurement Technology

- The higher resolution makes the measurement more precise compared with ultrasonic measurement.
- The longer wavelength provides greater penetration of eye tissue than other optical biometers, enabling precise measurement data.

## Placido Disc

- The 50-ring design Placido disc can provide more accurate analysis results.
- The large cone design covers more than 9.8mm diameter of central cornea so as to obtain more precise measurement data.

# Key Highlights of AL550

## Efficiency and Versatility

- Binocular biometry and topography examination **in 40 seconds** and binocular biometry examination **in 25 seconds**
- **Up to 23 parameters** can be obtained **in 1 measurement**
- Enables multi-scenario clinical applications

## Precision and Reliability

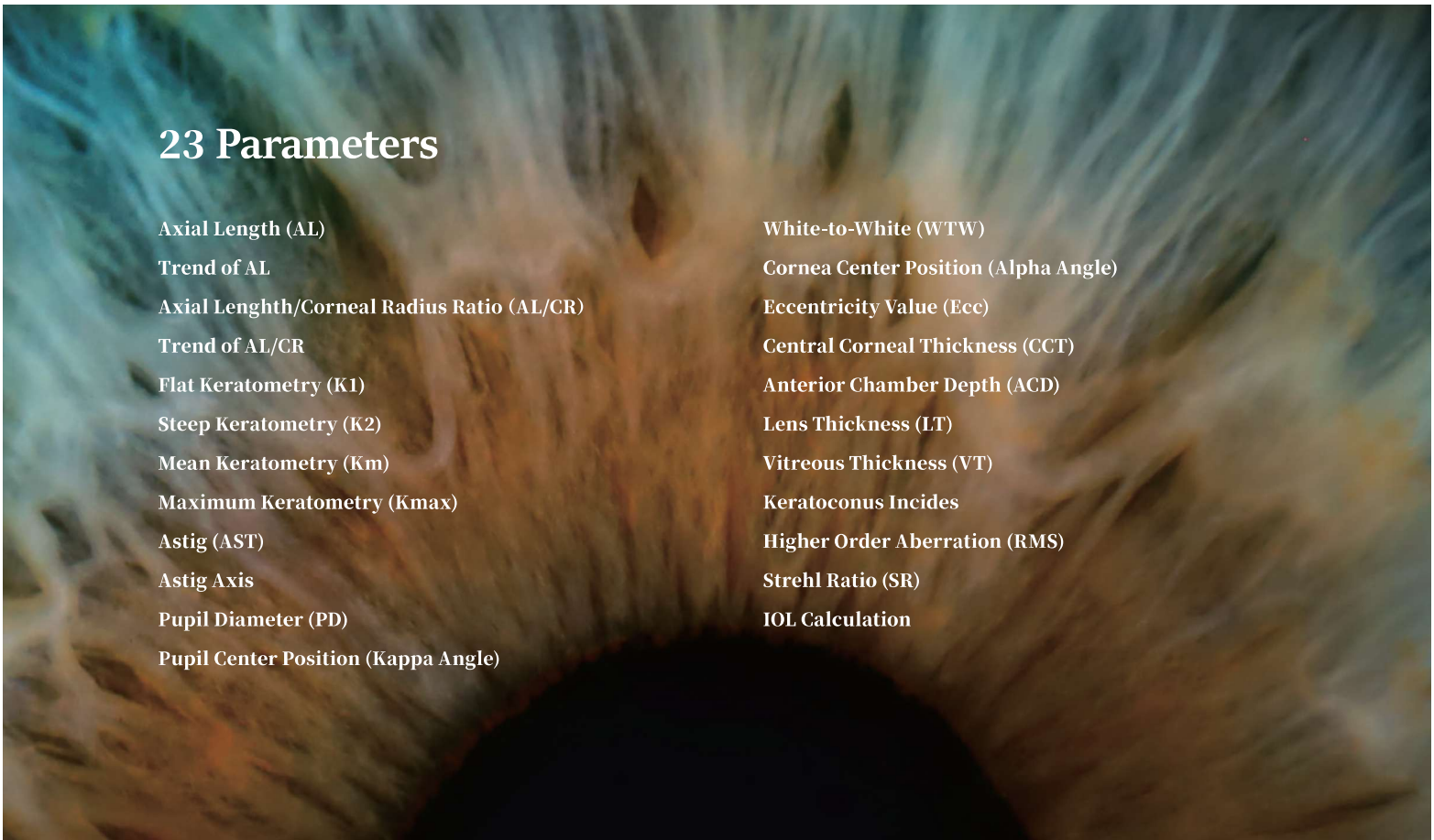
- **1,060nm wavelength** light source with strong penetrating power for accurate biometry measurement
- **125,600 data analysis points** for accurate anterior corneal surface measurement

## Compact Design and Ease of Operation

- **Space-saving design** for operation even in tight space
- **Selectable operation mode** and **rapid capture** for easier measurement process

## Safety and Comfort

- **Non-contact measurement** avoids the risk of cross-infection and corneal damage
- Topical anesthesia and pupil dilation are not required
- Under the guidance of **voice prompt**, patients are able to cooperate more smoothly and experience greater comfort



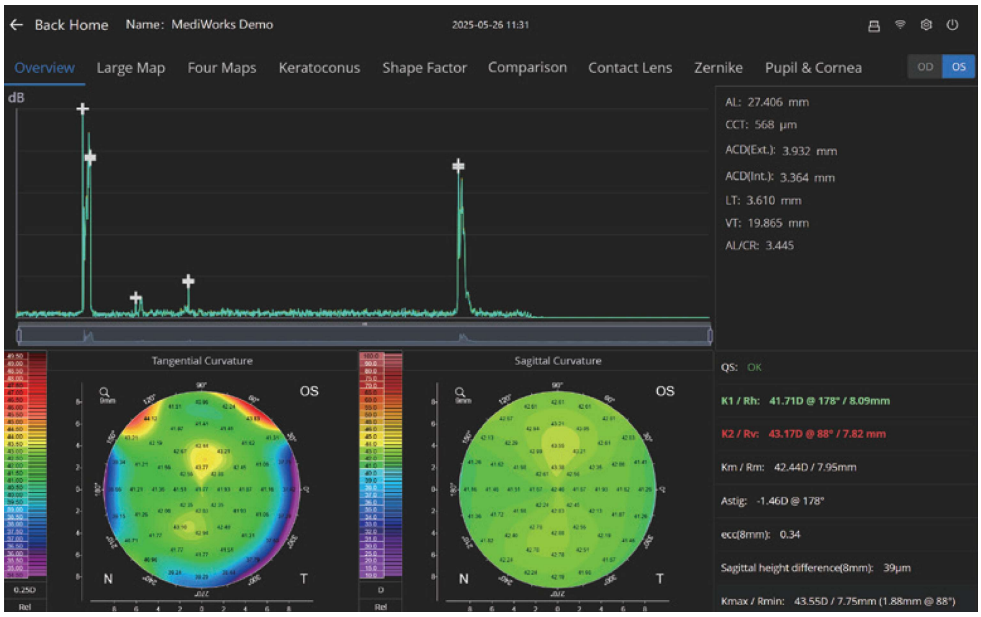
## 23 Parameters

**Axial Length (AL)**
**Trend of AL**
**Axial Lenghth/Corneal Radius Ratio (AL/CR)**
**Trend of AL/CR**
**Flat Keratometry (K1)**
**Steep Keratometry (K2)**
**Mean Keratometry (Km)**
**Maximum Keratometry (Kmax)**
**Astig (AST)**
**Astig Axis**
**Pupil Diameter (PD)**
**Pupil Center Position (Kappa Angle)**
**White-to-White (WTW)**
**Cornea Center Position (Alpha Angle)**
**Eccentricity Value (Ecc)**
**Central Corneal Thickness (CCT)**
**Anterior Chamber Depth (ACD)**
**Lens Thickness (LT)**
**Vitreous Thickness (VT)**
**Keratoconus Incides**
**Higher Order Aberration (RMS)**
**Strehl Ratio (SR)**
**IOL Calculation**

# Clinical Application in Cataract Surgery

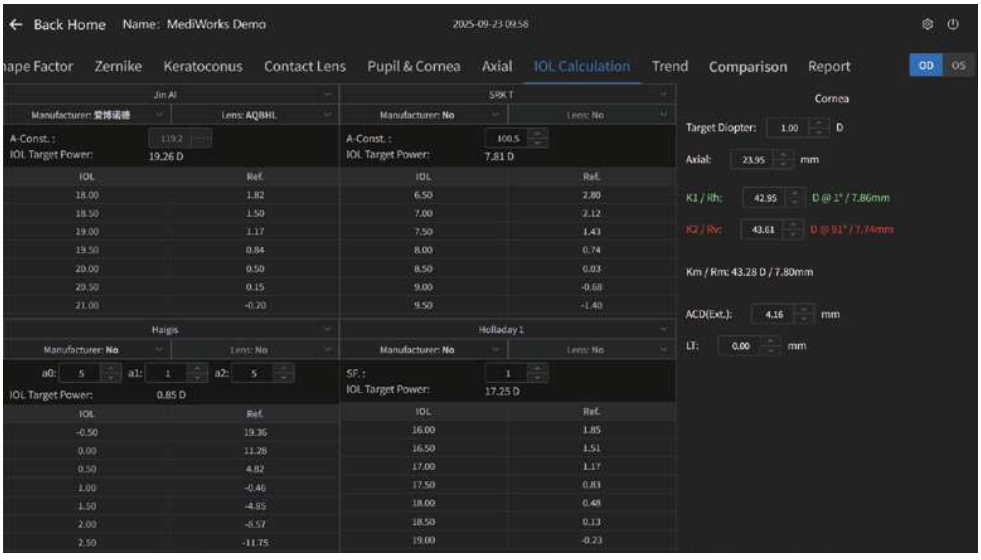
## Measurement of Axial Length, Corneal Curvature and Anterior Chamber Depth

AL551 provides accurate measurements such as axis length, corneal curvature and anterior chamber depth, which are crucial for the precise prediction of IOL power.



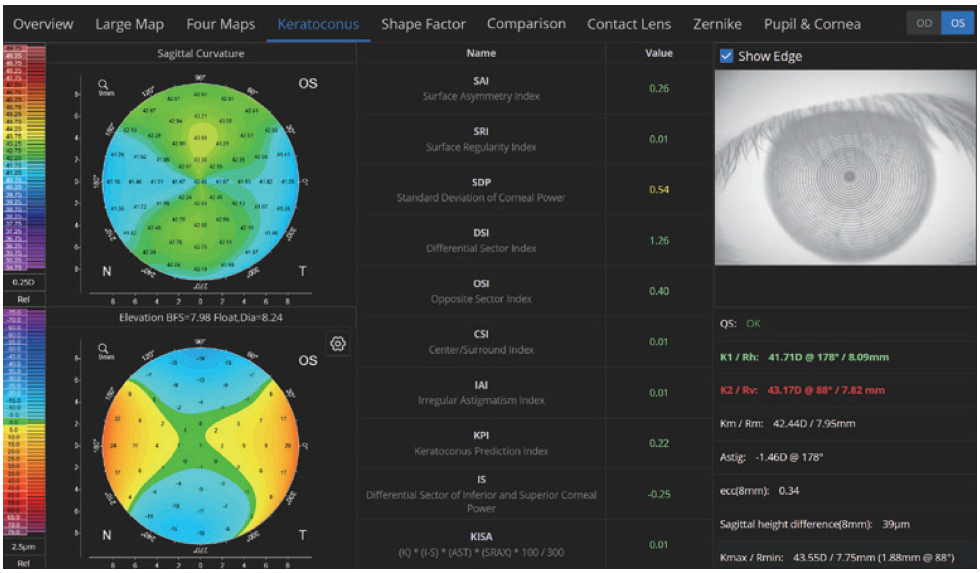

## IOL Calculation

AL550 integrates multiple IOL calculation formulas, including Pearl-DGS, Jin, SRK-T, Holladay 1, Hoffer Q and Haigis, enabling rapid calculation of IOL power for cataract patients. Moreover, a study found the Pearl-DGS and Jin formulas demonstrate prediction accuracy similar to that of the Barrett Universal II.



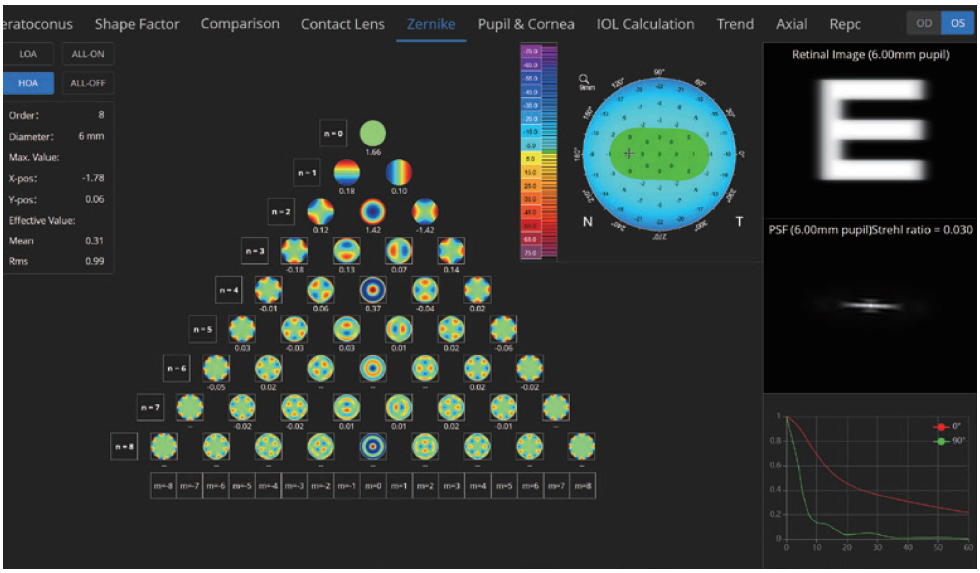
## Keratoconus Screening

AL550 provides the maps of anterior corneal sagittal curvature and elevation, and a variety of keratoconus indices, enabling eye-care professionals to intuitively and quickly understand the patient's corneal morphology and assisting in the early screening of diseases such as keratoconus and marginal corneal degeneration.




## Wavefront Aberration Analysis & Visual Quality Analysis

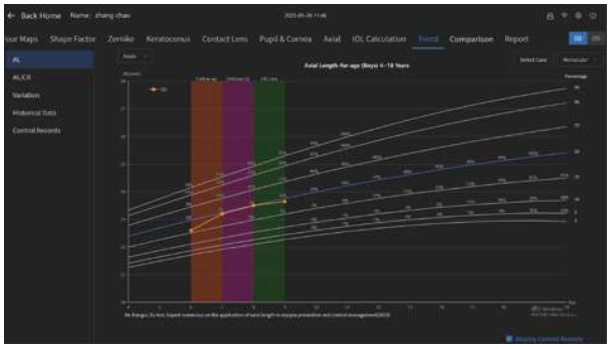
AL550 provides Zernike wavefront aberration data of the corneal anterior surface with up to 8th order and a maximum diameter of 12mm. Meanwhile, it can also simulate the visual quality image through more intuitive ways such as Retina Imaging, Point Spread Function (PSF, including Strehl ratio), and Modulation Transfer Function (MTF), helping improve the efficiency of eye-care professional-to-patient communication.



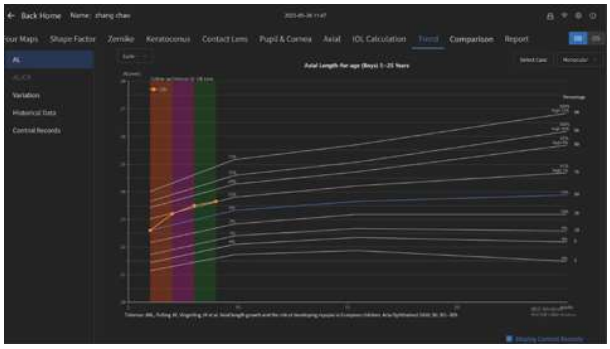
# Additional Clinical Applications

## Myopia Progression Monitoring

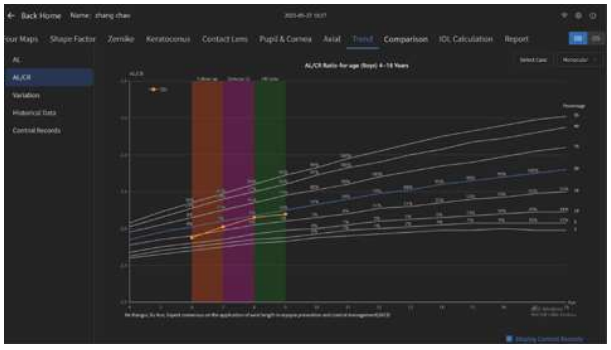
AL550 provides the trend curves of axial length, axial length/corneal radius ratio, axial length variation over time, and myopia & high myopia probability in the European and Asian population, and allows eye-care professionals to record myopia interventions so as to provide important evidence for monitoring myopia progression in children and adolescents, evaluating the effectiveness of current myopia interventions and predicting future myopia and high myopia risk.



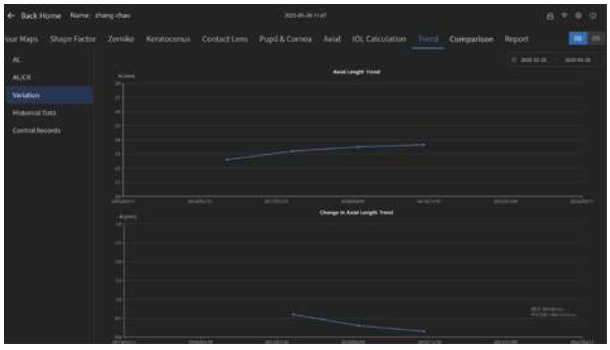
Trend Curve of AL, Myopia & High Myopia Probability in the European Population [1]



Trend Curve of AL & Myopia Probability in the Asian Population [2]



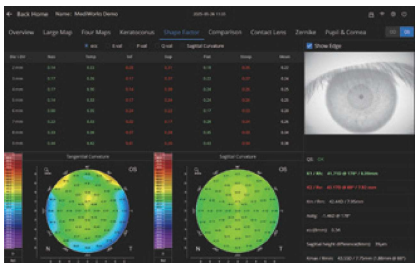
Trend Curve of AL/CR & Myopia Probability in the Asian Population [2]



Axial Length Variation over Time [2]

## Orthokeratology Lens Fitting

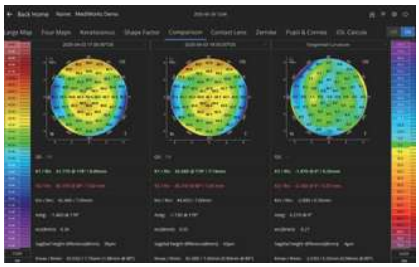
AL550 provides comprehensive orthokeratology lens fitting parameters (including flat K, steep K, astigmatism, flat eccentricity, steep eccentricity, and corneal sagittal height difference) for trial lens selection. It also offers the lens simulated fitting function to improve fitting efficiency and reduce patient discomfort, while generating tangential curvature difference map before and after lens wearing for effect evaluation.



Selection of Orthokeratology Trial Lens



Orthokeratology Lens Simulated Fitting



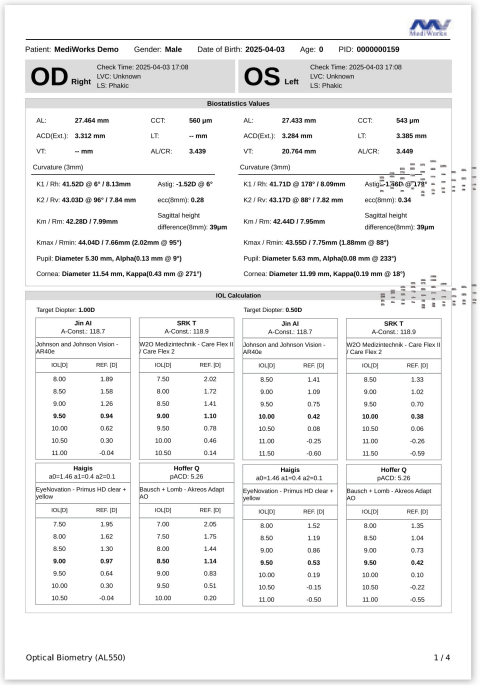
Evaluation of Orthokeratology Lens Wearing Effect


[1] Jan Willem Lodewijk Tideman, et al. Axial length growth and the risk of developing myopia in European children. ActaOphthalmol. 2018:96:301–309.

[2] He X, Sankaridurg P, Naduvilath T, et al. Normative data and percentile curves for axial length and axial length/corneal curvature in Chinese children and adolescents aged 4–18 years. Br J Ophthalmol. 2023;107:167–175.


AL 550

# Comprehensive and Easy-to-access Digital Report


Patient: MediWorks Demo Gender: Male Date of Birth: 2025-04-03 Age: 0 PID: 0000000159

OD Right — Check Time: 2025-04-03 17:08 LVC: Unknown LS: Phakic

OS Left — Check Time: 2025-04-03 17:08 LVC: Unknown LS: Phakic

Biostatistics Values

| OD | | | | OS | | | |
|---|---|---|---|---|---|---|---|
| AL: | 27.464 mm | CCT: | 560 μm | AL: | 27.433 mm | CCT: | 543 μm |
| ACD(Ext.): | 3.312 mm | LT: | -- mm | ACD(Ext.): | 3.284 mm | LT: | 3.385 mm |
| VT: | -- mm | AL/CR: | 3.439 | VT: | 20.764 mm | AL/CR: | 3.449 |

Curvature (3mm)

OD: K1 / Rh: 41.52D @ 6° / 8.13mm; Astig: -1.52D @ 6°; K2 / Rv: 43.03D @ 96° / 7.84 mm; ecc(8mm): 0.28; Km / Rm: 42.28D / 7.99mm; Sagittal height difference(8mm): 39μm; Kmax / Rmin: 44.04D / 7.66mm (2.02mm @ 95°); Pupil: Diameter 5.30 mm, Alpha(0.13 mm @ 9°); Cornea: Diameter 11.54 mm, Kappa(0.43 mm @ 271°)

OS: K1 / Rh: 41.71D @ 178° / 8.09mm; Astig: -1.46D @ 178°; K2 / Rv: 43.17D @ 88° / 7.82 mm; ecc(8mm): 0.34; Km / Rm: 42.44D / 7.95mm; Sagittal height difference(8mm): 39μm; Kmax / Rmin: 43.55D / 7.75mm (1.88mm @ 88°); Pupil: Diameter 5.63 mm, Alpha(0.08 mm @ 233°); Cornea: Diameter 11.99 mm, Kappa(0.19 mm @ 18°)

IOL Calculation

Target Diopter: 1.00D

| Jin AI A-Const.: 118.7 | | SRK T A-Const.: 118.9 | |
|---|---|---|---|
| Johnson and Johnson Vision - AR40e | | W2O Medizintechnik - Care Flex II / Care Flex 2 | |
| IOL[D] | REF. [D] | IOL[D] | REF. [D] |
| 8.00 | 1.89 | 7.50 | 2.02 |
| 8.50 | 1.58 | 8.00 | 1.72 |
| 9.00 | 1.26 | 8.50 | 1.41 |
| **9.50** | **0.94** | **9.00** | **1.10** |
| 10.00 | 0.62 | 9.50 | 0.78 |
| 10.50 | 0.30 | 10.00 | 0.46 |
| 11.00 | -0.04 | 10.50 | 0.14 |

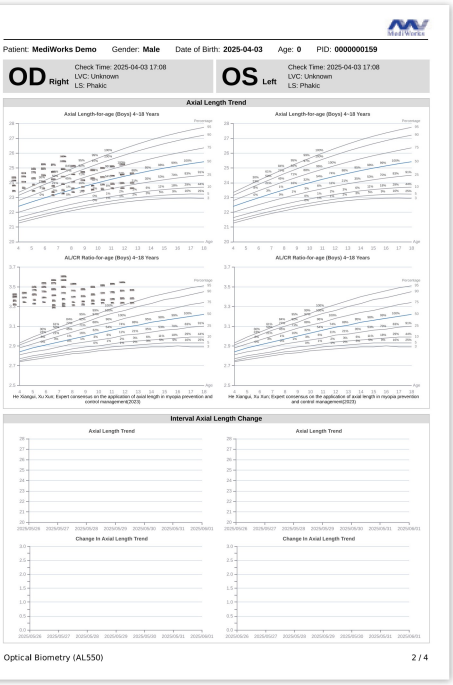
| Haigis a0=1.46 a1=0.4 a2=0.1 | | Hoffer Q pACD: 5.26 | |
|---|---|---|---|
| EyeNovation - Primus HD clear + yellow | | Bausch + Lomb - Akreos Adapt AO | |
| IOL[D] | REF. [D] | IOL[D] | REF. [D] |
| 7.50 | 1.95 | 7.00 | 2.05 |
| 8.00 | 1.62 | 7.50 | 1.75 |
| 8.50 | 1.30 | 8.00 | 1.44 |
| **9.00** | **0.97** | **8.50** | **1.14** |
| 9.50 | 0.64 | 9.00 | 0.83 |
| 10.00 | 0.30 | 9.50 | 0.51 |
| 10.50 | -0.04 | 10.00 | 0.20 |

Target Diopter: 0.50D

| Jin AI A-Const.: 118.7 | | SRK T A-Const.: 118.9 | |
|---|---|---|---|
| Johnson and Johnson Vision - AR40e | | W2O Medizintechnik - Care Flex II / Care Flex 2 | |
| IOL[D] | REF. [D] | IOL[D] | REF. [D] |
| 8.50 | 1.41 | 8.50 | 1.33 |
| 9.00 | 1.09 | 9.00 | 1.02 |
| 9.50 | 0.75 | 9.50 | 0.70 |
| **10.00** | **0.42** | **10.00** | **0.38** |
| 10.50 | 0.08 | 10.50 | 0.06 |
| 11.00 | -0.25 | 11.00 | -0.26 |
| 11.50 | -0.60 | 11.50 | -0.59 |

| Haigis a0=1.46 a1=0.4 a2=0.1 | | Hoffer Q pACD: 5.26 | |
|---|---|---|---|
| EyeNovation - Primus HD clear + yellow | | Bausch + Lomb - Akreos Adapt AO | |
| IOL[D] | REF. [D] | IOL[D] | REF. [D] |
| 8.00 | 1.52 | 8.00 | 1.35 |
| 8.50 | 1.19 | 8.50 | 1.04 |
| 9.00 | 0.86 | 9.00 | 0.73 |
| **9.50** | **0.53** | **9.50** | **0.42** |
| 10.00 | 0.19 | 10.00 | 0.10 |
| 10.50 | -0.15 | 10.50 | -0.22 |
| 11.00 | -0.50 | 11.00 | -0.55 |

Optical Biometry (AL550) 1 / 4


Patient: MediWorks Demo Gender: Male Date of Birth: 2025-04-03 Age: 0 PID: 0000000159

OD Right — Check Time: 2025-04-03 17:08 LVC: Unknown LS: Phakic

OS Left — Check Time: 2025-04-03 17:08 LVC: Unknown LS: Phakic

Axial Length Trend

Axial Length-for-age (Boys) 4~18 Years

AL/CR Ratio-for-age (Boys) 4~18 Years

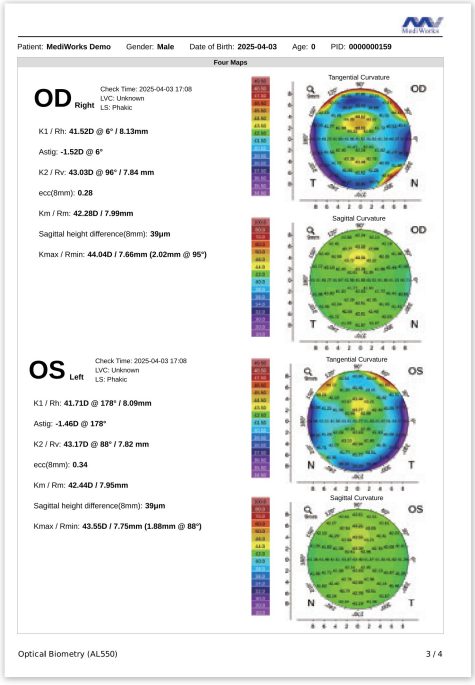
He Xiangjie, Xu Xue. Expert consensus on the application of axial length in myopia prevention and control management[2023].

Interval Axial Length Change

Axial Length Trend

Change In Axial Length Trend

Optical Biometry (AL550) 2 / 4


Patient: MediWorks Demo Gender: Male Date of Birth: 2025-04-03 Age: 0 PID: 0000000159

Four Maps

OD Right — Check Time: 2025-04-03 17:08 LVC: Unknown LS: Phakic

K1 / Rh: 41.52D @ 6° / 8.13mm

Astig: -1.52D @ 6°

K2 / Rv: 43.03D @ 96° / 7.84 mm

ecc(8mm): 0.28

Km / Rm: 42.28D / 7.99mm

Sagittal height difference(8mm): 39μm

Kmax / Rmin: 44.04D / 7.66mm (2.02mm @ 95°)

Tangential Curvature

Sagittal Curvature

OS Left — Check Time: 2025-04-03 17:08 LVC: Unknown LS: Phakic

K1 / Rh: 41.71D @ 178° / 8.09mm

Astig: -1.46D @ 178°

K2 / Rv: 43.17D @ 88° / 7.82 mm

ecc(8mm): 0.34

Km / Rm: 42.44D / 7.95mm

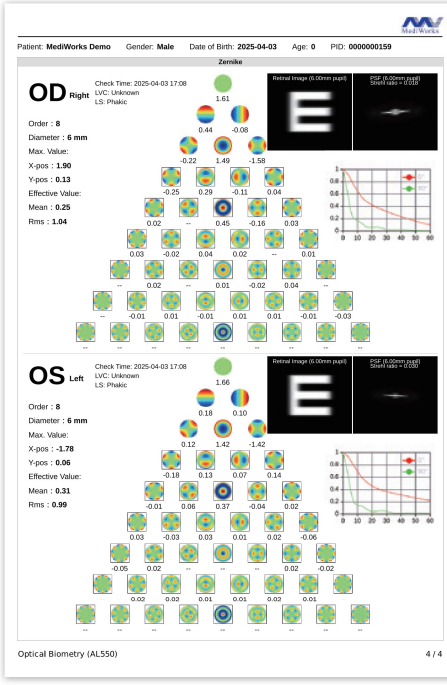
Sagittal height difference(8mm): 39μm

Kmax / Rmin: 43.55D / 7.75mm (1.88mm @ 88°)

Tangential Curvature

Sagittal Curvature

Optical Biometry (AL550) 3 / 4


Patient: MediWorks Demo Gender: Male Date of Birth: 2025-04-03 Age: 0 PID: 0000000159

Zernike

OD Right — Check Time: 2025-04-03 17:08 LVC: Unknown LS: Phakic

Order : 8
Diameter : 6 mm
Max. Value:
X-pos : 1.90
Y-pos : 0.13
Effective Value:
Mean : 0.25
Rms : 1.04

Retinal Image (6.00mm pupil)

PSF (6.00mm pupil) Strehl ratio = 0.010

OS Left — Check Time: 2025-04-03 17:08 LVC: Unknown LS: Phakic

Order : 8
Diameter : 6 mm
Max. Value:
X-pos : -1.78
Y-pos : 0.06
Effective Value:
Mean : 0.31
Rms : 0.99

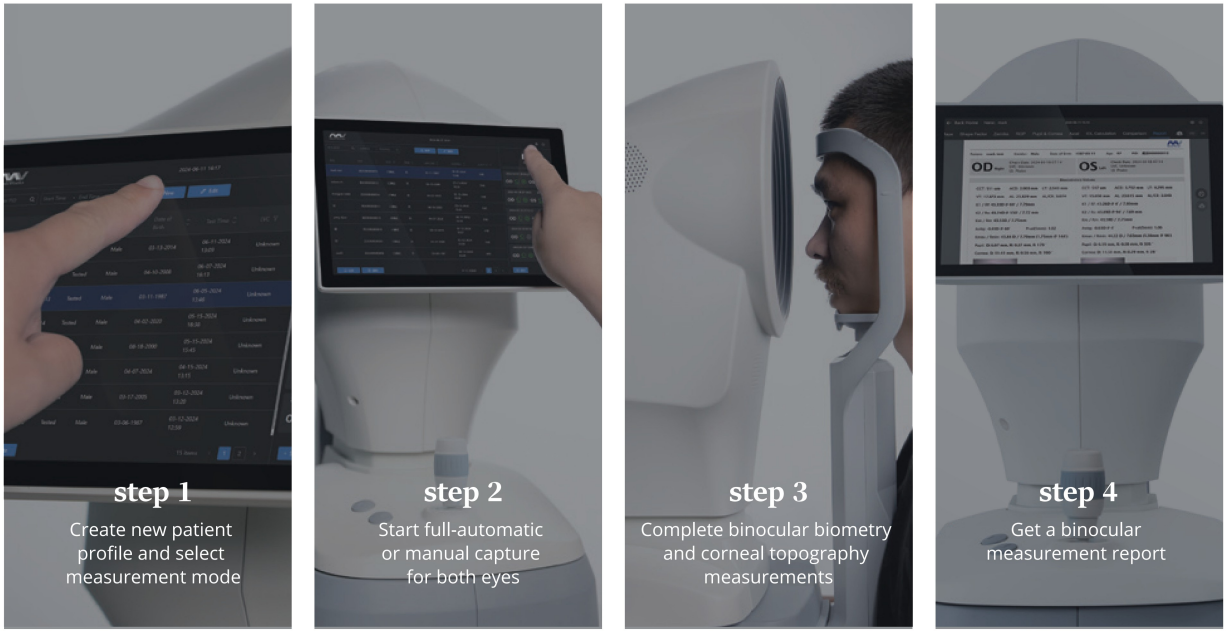
Retinal Image (6.00mm pupil)

PSF (6.00mm pupil) Strehl ratio = 0.030

Optical Biometry (AL550) 4 / 4

A comprehensive report with rich information on ocular biometric data, IOL calculations, corneal topographic maps and wave front aberrations of anterior corneal surface can help eye-care professionals scientifically evaluate patients' eye health status, so as to better guide their daily practices on cataract surgery.

# Ease of Operation




User-friendly operation and fast image capture realizes a super fast examination process.

# Integrated Compact Design

The integrated design of high-definition touch LCD screen and measuring device greatly saves the occupied space of diagnosis room.

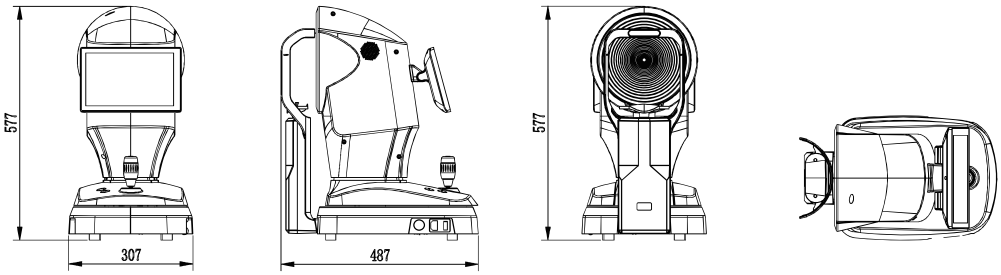


# Specifications

| FEATURE | SPECIFICATION |
|---|---|
| Keratoscopic cone | 50-ring Placido |
| Analyzed points | 125,600 |
| Measured points | 18,000 |
| Corneal coverage | 9.8mm |
| Diopter power range | 32.14 ~ 61.36D (±0.1D) |
| Axial Biometry | Optical Low-coherence Reflectometry (1,060nm ) |
| Focus mode | One-touch,XYZ axes autofocus tracking measurement |
| OS/OD Recognition | Automatic |
| Monitor | 10.1" touchscreen |
| Illumination | Red, Infrared |
| Database | Internal |
| Working environment | +5°C~+40°C, Relative humidity ≤80% (no condensing), |
| | Atmospheric pressure 800 – 1,060 hPa |
| Power supply | ~100 - 240 VAC, 50/60 HZ |
| Power consumption | 100 VA |
| Dimensions | 307 mm (W) × 487 mm (L) × 577 mm (H) |
| Weight | 21kg |
| Printing options | USB printer, Network printer, PDF on network shared folder, |
| | PDF on USB PDF or Image on network folder or on USB |
| Operating System | Linux |
| Hard Disk | 1TB |

## INFORMATION ON MEASUREMENTS

| MEASUREMENT | Measuring Range | Tolerance Value | Display Resolution |
|---|---|---|---|
| Axlal Length | 0 - 40 mm | ±10 μm | 1 μm |
| Corneal Thickness | 0.2 - 1.2 mm | ±10 μm | 1 μm |
| Anterior Chamber Depth | 0.7 - 8 mm | ±10 μm | 1 μm |
| Lens Thickness | 1.5 - 6.5 mm | ±10 μm | 1 μm |
| White-to-White | 8 - 16 mm | ±0.1 mm | |
| Pupil Diameter | 1 - 13 mm | | |
| Projection ring diameter | > 9.8 mm | | |
| Corneal Radius of Curvature | 5.5 - 10.5 mm (Accuracy ± 0.02 mm) | | |
| Axis | 0 ~ 180° | | |




CE NMPA FDA MDSAP ISO

Shanghai MediWorks Precision Instruments Co.,Ltd.

Add: Building 7, Ming Pu Plaza, No. 3279, San Lu Rd, Min Hang District, Shanghai, 201100, China
Tel: +86-21-54260421 54260423
Email: marketing@mediworks.biz

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www.mediworks.biz

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**Details** Make the **Difference**